What SHMtrace gives you, and how to read it

You provide

- **reads per animal** — any platform
- **AID-null controls** — two or more

+ a sample sheet

SHMtrace — calls and tests against the controls

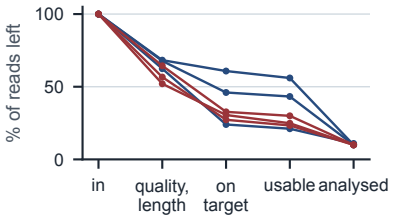
You get

- **report.html** — every result on one page
- **figures** — PNG, SVG or PDF
- **tables** — Excel and CSV, every number
- **{ } project.json** — reopens in the app

Examples redrawn from one run (JH4 intron, nanopore, 500 reads per mouse): wild type · AID-null (*Aicda−/−*)

1 Did enough reads get through?

qc_waterfall



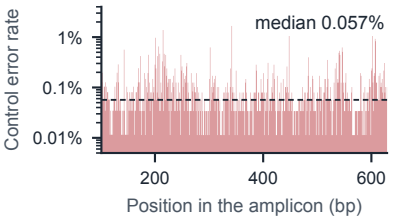
One line per library: the share of reads left after each filter. The last step is the depth you set, not a loss.

Watch for: a library far below the others.

Also: coverage · quality_histogram · saturation

2 Are the calls real?

control_background



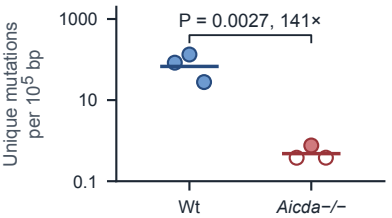
The error rate the AID-null controls showed at each position. A mutation is called only if it clears this floor, with predicted false calls held under 5%.

Watch for: spikes: calls there need much more support.

Also: volcano · single_read_excess · replicate_concordance

3 How much mutation?

density_by_group



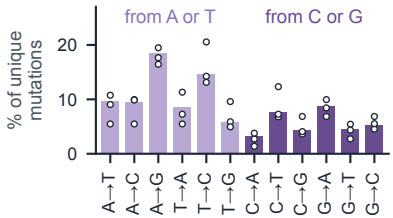
One dot per animal (open: none found); bars are geometric means. The test counts each animal once, however many reads it gave.

Watch for: frequency counts a clone's mutations in every read; unique density counts them once.

Also: mutation_frequency_by_group · mutated_reads_by_group · mutation_load · combined_density · indels

4 What kind of mutation?

spectrum_by_group



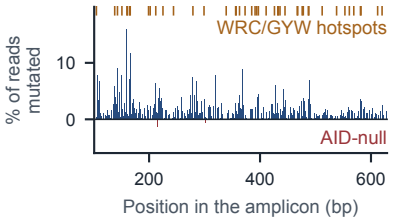
The 12 base changes: bars for wild type, dots per mouse. AID leaves C→T and G→A at WRC/GYW motifs (54% of G:C mutations here).

Watch for: compare amplicons on the composition-corrected spectrum.

Also: substitution_matrix · hotspot_targeting · spectrum_corrected · targeting_spectrum · mutability_hedgehog

5 Where along the amplicon?

positional_map



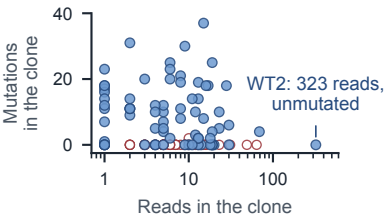
The share of reads mutated at each position: wild type up, AID-null down. Peaks on hotspot ticks are the signature of AID.

Watch for: a peak in the controls too: a germline variant or an artefact.

Also: positional_by_sample · targeting_profile

6 Which clones carry them?

clone_map



Clones from the V(D)J junction; each dot is one clone. In WT2, one unmutated clone held 81% of reads.

Watch for: one clone repeating its mutations in many reads.

Also: clonality · phasing_blocks

Other routes: a single sample from the command line writes spectrum, spatial and mutation_load; the legacy compare writes three more.